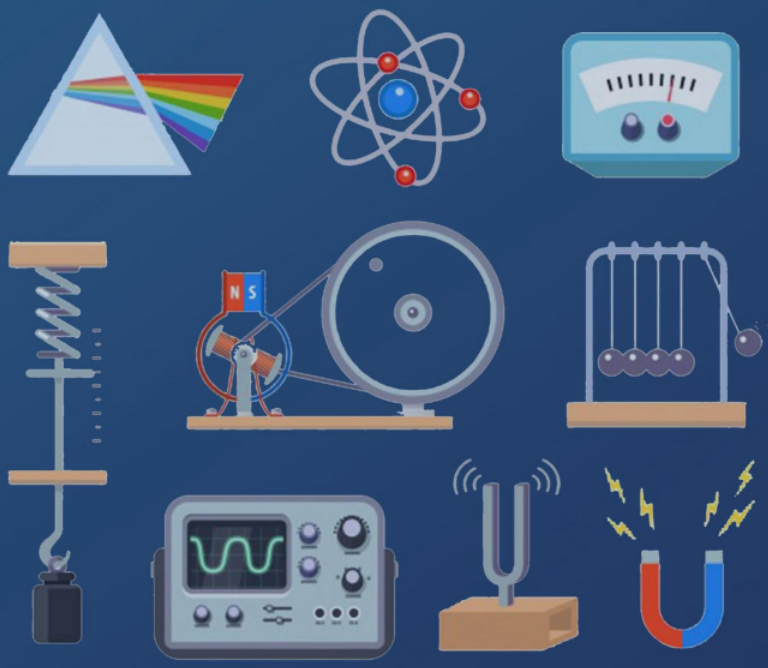
PHYSICS

Communication System



SHEET

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Note

Detailed solutions are available on the eSaral App.

Module Description



For mastering the concepts only learning is not sufficient. We have to practice and apply those concepts in problem solving. This sheet does just that. It contains a collection of problems segregated in the following exercises to help you master the concepts in a systematic and organized way.

"Practice makes a man perfect"

1. Concept builder – 1 & 1A

As soon as you have finished learning the concept do the problems from these exercises first. These exercises contains easy level questions to help you build your concepts.

1 ⟶ Contains Single Correct Type questions

1A ⟶ Contains pattern based questions incorporating the latest JEE Advanced based patterns like more than one correct, matching list, match the column, etc.

2. Brain Booster – 2 & 2A

Now that you have built your concepts it's time to master them by solving Brain Boosting problems. Don't hurry through these problems. Take time to solve & learn from them. These exercises contains Medium & Tough level problems.

Do questions from 2 & 2A after attempting 1 & 1A

2 ⟶ Single Correct Type questions.

2A ⟶ Pattern Based questions.

3. Simulator – JM & JA

Contains questions from previous year JEE Mains & JEE Advanced questions in exercise JM & exercise JA respectively. Get the real taste & feel of the type of questions being asked in JEE. It's a great tool for simulating your mind with JEE problems.

These exercises are not included in the module but are provided separately.

JM ⟶ JEE Mains previous years topic wise questions.

JA ⟶ JEE Advance previous years topic wise questions.

Home Work Index

Problem solving is an integral part of learning.

Find questions to solve after each video in the homework Index. Make sure that you attempt all the problems (in Ex 1 to 2A) after learning a topic from the videos. For example if you have finished topic 8, first attempt all the problems listed in the index corresponding to topic 8 before proceeding to the video of topic 9.

Sr.No.	Topic Name	Ex-1	Ex-2
1	Introduction	1,2	
2	Propagation of Waves	3-9	1,2
3	Modulation and its types	10,11	3-5
4	Amplitude Modulation	12-14	6-13

Exercise - 1 Concept Builder

SINGLE CORRECT TYPE

*Each question has FOUR options for correct answer. **ONLY ONE** of these four options is correct option. For each question, choose the correct option to answer the question.*

Introduction

Q1. A basic communication system consists of
(A) transmitter
(B) information source.
(C) user of information
. (D) channel
(E) receiver.
Choose the correct sequence in which these are arranged in a basic communication system :
(A) ABCDE (B) BADEC
(C) BDACE (D) BEADC

Q2. The process of superimposing signal frequency (i.e. audio wave) on the carrier wave is known as :
(A) Transmission (B) Reception
(C) Modulation (D) Detection

Propagation of waves

Q3. Frequencies in the UHF range normally propagate by means of :-
(A) Ground waves (B) Sky waves
(C) Satellite waves (D) Space waves

Q4. Long distance short-wave radio broad-casting generally uses :-
(A) Ground wave
(B) Ionospheric wave
(C) Direct wave
(D) Sky wave

Q5. For television broadcasting, the frequency employed is normally :-
(A) 30 – 300 MHz
(B) 30 – 300 GHz
(C) 30 – 300 KHz
(D) 30 – 300 Hz

Q6. In short wave communication waves of which of the following frequencies will be reflected back by the ionospheric layer having electron density 10^{11} m^{-3} ?
(A) 2 MHz
(B) 10 MHz
(C) 12 MHz
(D) 18 MHz

Q7. Range of frequencies allotted for commercial FM radio broadcast is :-
(A) 88 to 108 MHz
(B) 88 to 108 kHz
(C) 8 to 88 MHz
(D) 88 to 108 GHz

Q8. The maximum distance upto which TV transmission from a TV tower of height h can be received is proportional to :-
(1) $h^{1/2}$ (2) h
(3) $h^{3/2}$ (4) h^2

Q9. For skywave propagation of a 10 MHz signal, what should be the minimum electron density in ionosphere –
(A) ~ $1{\cdot}2 \times 10^{12}$ m^{-3}
(B) ~ 10^6 m^{-3}
(C) ~ 10^{14} m^{-3}
(D) ~ 10^{22} m^{-3}

Modulation and its types

Q10. In amplitude modulation :-

(A) Only the amplitude is changed but frequency remains same

(B) Both the amplitude and frequency change equally

(C) Both the amplitude and frequency change unequally

(D) None the these

Q11. In frequency modulation :-

(A) The amplitude of career wave varies according to the frequency of message signal

(B) The frequency of career wave varies according to the amplitude of message signal

(C) The frequency of career wave varies according to the frequency of message signal

(D) The amplitude of career wave varies according to the amplitude of message signal

Amplitude Modulation

Q12. AM is used for broadcasting because :-

(A) It is more noise immune than other modulation systems

(B) It requires less transmitting power compared with other systems

(C) Its use avoids receiver complexity

(D) No other modulation system can provide faithful transmission for the necessary bandwidth

Q13. In an amplitude modulated wave for audio-frequency of 500 cycles/second, the appropriate carrier frequency will be :-

(A) 50 cycles/sec.

(B) 100 cycles/sec.

(C) 500 cycles/sec

(D) 50,000 cycles/sec.

Q14. An antenna is of height 500 m. What will be its range (Radius of the earth is 6400 km)?

(A) 800 km (B) 100 km

(C) 50 km (D) 80 km

Exercise - 2 *Brain Booster*

SINGLE CORRECT TYPE

Each question has FOUR options for correct answer. ***ONLY ONE*** *of these four options is correct option. For each question, choose the correct option to answer the question.*

Q1. The maximum range of ground or surface wave propagation depends on :-

(A) The frequency of the radio waves only

(B) Power of the transmitter only

(C) Both of them

(D) None of them

Q2. Three waves A, B and C of frequencies 1600 kHz, 5 MHz and 60 MHz, respectively are to be transmitted from one place to another. Which of the following is the most appropriate mode of communication :

(A) A is transmitted via space wave while B and C are transmitted via sky wave.

(B) A is transmitted via ground wave, B via sky wave and C via space wave.

(C) B and C are transmitted via ground wave while A is transmitted via sky wave.

(D) B is transmitted via ground wave we while A and C are transmitted via space wave.

Q3. If a carrier wave of 1000 kHz is used to carry the signal, the length of transmitting antenna will be equal to :-

(A) 3 m (B) 30m

(C) 75 m (D) 3000 m

Q4. **Statement-1 :** An audio signal is transmitted over long distances by translating it into high frequency wave.

Statement-2 : Size of transmitting antenna is to be at least $\lambda/4$ in dimension.

(A) Statement-1 is true, statement-2 is true and statement-1 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation of statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q5. Modulation factor determines :-

(A) Only the strength of the transmitted signal

(B) Only the quality of the transmitted signal

(C) Both the strength and quality of the signal

(D) None of the above

Q6. Which of the following is/are the limitations of amplitude modulation ?

(A) Clear reception

(B) High efficiency

(C) Small operating range

(D) Good audio quality

Q7. For a carrier frequency of 100 kHz and a modulating frequency of 5 kHz what is the width of AM transmission :-

(A) 5 kHz (B) 10kHz

(C) 20 kHz (D) 200 KHz

Q8. A speech signal of 3kHz is used to modulate a carrier signal of frequency 1 MHz, using amplitude modulation. The frequencies of the side bands will be :

(A) 1.003 MHz and 0.997 MHz.

(B) 3001 kHz and 2997 kHz.

(C) 1003 kHz and 1000 kHz.

(D) 1 MHz and 0.997 MHz.

Q9. A message signal of frequency ω_m is superposed on a carrier wave of frequency ω_c to get an amplitude mediated wave (AM). The frequency of the AM wave will be :

(A) ω_m (B) ω_c

(C) $\frac{\omega_c + \omega_m}{2}$ (D) $\frac{\omega_c - \omega_m}{2}$

Q10. Identify the mathematical expression for amplitude modulated wave :

(A) $A_c \sin [\{\omega_c + kv_m(t)\}t + \phi]$

(B) $A_c \sin \{\omega_c t + \phi + k_2 v_m(t)\}$

(C) $\{A_c + kv_m(t)\} \sin (\omega_c t + \phi)$

(D) $A_c + v_m(t) \sin (\omega_c t + \phi)$

Q11. $C(t)$ and $m(t)$ are used to generate an AM signal. The modulation index of generated AM signal is 0.5.

Then the quantity $\frac{P_{\text{Total SB}}}{P_{\text{Carrier}}} =$

(A) 1/8 (B) 1/4 (C) 2/3 (D) 9/8

Q12. A message signal $m(t) = 4\cos 2000\pi t$ modulates the carrier $C(t) = \cos 2\pi f_c t$ where $f_c = 1\text{MHz}$ to produce an AM signal. For demodulation using envelope detector the time constant RC should satisfy

(A) $0.5\text{ms} < RC < 1\text{ms}$

(B) $1\mu s \leq RC \leq 0.5\text{ms}$

(C) $RC \leq 1\mu s$

(D) $RC \geq 1\mu s$

Q13. The antenna current of an AM transmitter is 8 *A* when only carrier is sent but increases to 8.96 *A* when the carrier is modulated sinusoidally. The percentage modulation is:

(A) 50% (B) 60% (C) 65% (D) 71%

Answer Key

Ex-1									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
B	C	D	B	A	A	A	A	A	A
11	**12**	**13**	**14**						
B	C	D	D						

Ex-2									
1	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**
C	B	C	A	C	C	B	A	B	C
11	**12**	**13**							
A	B	D							